

Adiabatic non-Abelian braiding of imperfect Majorana bound states

Maximilian Nitsch¹, Viktor Svensson², William Samuelson¹, Konstantin Nestmann¹, Jeroen Danon³,
 Karsten Flensberg⁴, Rubén Seoane Souto^{5,6} and Martin Leijnse¹

¹*Division of Solid State Physics and NanoLund, Lund University, Lund, Sweden*

²*Department of Physics, University of Oslo, Oslo, Norway*

³*Department of Physics, Norwegian University of Science and Technology, Trondheim, Norway*

⁴*Center for Quantum Devices, Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark*

⁵*Departamento de Física Teórica de la Materia Condensada, Condensed Matter Physics Center (IFIMAC),
 and Instituto Nicolás Cabrera, Universidad Autónoma de Madrid, Madrid, Spain*

⁶*Quantum Advanced Research Center (QuARC), Consejo Superior de Investigaciones Científicas (CSIC),
 Sor Juana Inés de la Cruz 3, 28049 Madrid, Spain*



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Demonstration of a nontrivial result of quasiparticle exchange (or braiding) is usually considered the definitive proof of a topological phase with non-Abelian excitations, such as Majorana bound states (MBSs). However, in finite systems with disorder and smooth potential variations, the MBSs are imperfect in the sense that they are not fully isolated in space and can, to a varying degree, resemble conventional fermions. The main goal of this work is to reveal the braiding properties of isolated MBSs, regular fermions, and anything in between. We start from an experimentally attractive braiding protocol and find a way to compensate for the undesired splitting of the ground-state degeneracy which occurs during the protocol for imperfect MBSs. This leads to a braiding outcome that depends on the degree of MBS isolation but remains robust and non-Abelian except in the perfect fermion limit. Our protocol could be implemented in different platforms with non-Abelian excitations, including quantum-dot-based minimal Kitaev chains.

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Introduction. All elementary particles are either fermions or bosons, distinguished by whether particle exchange results in a sign change in the wave function or not. Quasiparticles in lower dimensions can, in principle, satisfy more general exchange rules. Non-Abelian anyons are the most exotic example, with particle exchange (or braiding) described by a nontrivial operator acting on the multiparticle wave function [1]. Direct experimental demonstration of non-Abelian braiding remains one of the main open challenges in modern physics [2,3] and has potential applications in error-protected topological quantum computing [1,4–6].

Majorana bound states (MBSs) [7–10], zero-energy excitations in one-dimensional topological superconductors, arguably provide the experimentally most accessible platform to demonstrate non-Abelian braiding (see, for example, Refs. [2,11–19]). Unfortunately, progress has been slowed by difficulties to unambiguously distinguish between MBSs and trivial states.

The approach to realize MBSs by building up artificial Kitaev chains [20] from quantum dots coupled via superconductors [21–26] has seen recent rapid experimental progress [27–34]. However, because these chains are typically rather

short, these MBSs lack true topological protection [22]. Non-Abelian braiding can still be possible with such MBSs, but requires fine-tuning [35–38] and is susceptible to noise [39]. In fact, the situation might be similar for supposedly topological realizations due to finite size effects, disorder, and smooth confining potentials [40–44]. In all these cases, the low-energy states can be described as MBSs that are imperfect in the sense that they are not fully isolated in space. On the other hand, a conventional zero-energy fermion can always be represented as two fully overlapping MBSs. Therefore, there is no strict border between non-Abelian MBSs and conventional fermions but rather a smooth transition between these two limiting cases.

In this Letter, we investigate the braiding properties in a model that interpolates between the MBS and fermion limits. We focus on a specific well-studied and experimentally attractive braiding protocol based on cyclically coupling three systems [14,15,17,19,35,36,45], one possible realization being three double quantum dots coupled via narrow superconducting regions as shown in Fig. 1(a). Each system is described by a model that hosts zero-energy MBSs that can be continuously tuned from being isolated (perfect MBS limit) to fully overlapping (conventional fermion limit).

Even though each isolated system hosts a zero-energy state, deviations from fully isolated MBSs lead to a breaking of ground-state degeneracy when the systems are coupled during the protocol. Even a small induced ground-state splitting gives rise to a dynamical phase that destroys the stable braiding result, while a large splitting (compared to the inverse protocol

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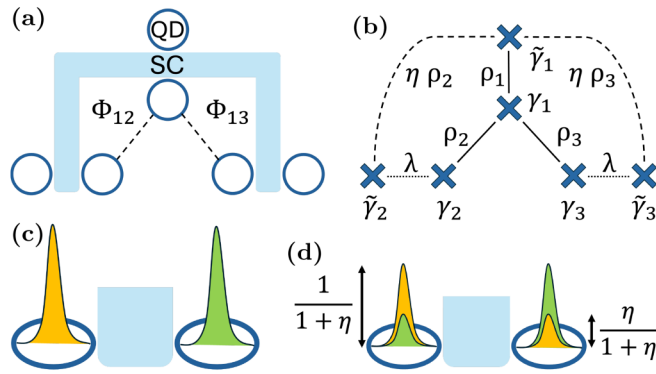


FIG. 1. (a) Example setup to implement coupling-based braiding in a system hosting MBSs in three minimal Kitaev chains based on quantum dots (QDs) connected via a superconductor (SC). Magnetic fluxes Φ_{12} , Φ_{13} control the relative phases of the SC-QD couplings. (b) Representation of the different MBS couplings used in Eq. (1). (c), (d) Cartoon of isolated MBSs (c) and spatially overlapping MBSs (d).

time) forces the system to always stay in the unique global ground state and gives rise to a trivial braiding result. To reveal the fundamental adiabatic exchange properties of imperfect MBSs, we develop a way to compensate for this splitting to maintain perfect degeneracy throughout the braiding protocol. The resulting protocol performs a braiding operation that features a non-Abelian result that is stable with respect to protocol speed and coupling strengths. Nevertheless, the result, in general, deviates from that of braiding isolated MBSs and converges to the trivial fermion result for fully overlapping MBSs. We present an analytical solution for the non-Abelian Berry phase that determines the braiding result in the adiabatic and degenerate limit and solve the general case numerically.

Model. The braiding is realized by connecting three systems, see Fig. 1(b), where couplings (lines) between different MBSs (crosses) are switched on and off [14,15,17,35,36,45]. A minimal representation is given by the Hamiltonian

$$\begin{aligned} H/\Delta = & \rho_1 i\gamma_1 \tilde{\gamma}_1 + \rho_2 i\gamma_1 \gamma_2 + \rho_3 i\gamma_1 \gamma_3 \\ & + \eta (\rho_2 i\tilde{\gamma}_1 \tilde{\gamma}_2 + \rho_3 i\tilde{\gamma}_1 \tilde{\gamma}_3) \\ & + \lambda (i\gamma_2 \tilde{\gamma}_2 + i\gamma_3 \tilde{\gamma}_3). \end{aligned} \quad (1)$$

The model includes three pairs of MBSs, γ_k , $\tilde{\gamma}_k$. Each pair encodes one fermionic state with two parity states $i\gamma_k \tilde{\gamma}_k = \pm 1$, thus yielding an eight-dimensional Hilbert space for the total system. Each term in Eq. (1) corresponds to a coupling between two MBSs (e.g., $\rho_1 i\gamma_1 \tilde{\gamma}_1$), assigning a certain energy ($\propto \pm \rho_1$) to each parity ($i\gamma_1 \tilde{\gamma}_1 = \pm 1$). A graphical representation of all included couplings is shown in Fig. 1(b). Total parity $\sigma = \prod_k i\gamma_k \tilde{\gamma}_k$ commutes with Eq. (1) and is therefore conserved. One possible effective implementation of this Hamiltonian can be achieved via three minimal Kitaev chains [22] arranged in a T junction—see Fig. 1(a) and Supplemental Material (SM) [46].

The first line of Eq. (1) includes all desired couplings needed for the braiding protocol. These are the only couplings that occur if each system hosts MBSs that are fully isolated in space—see Fig. 1(c). Couplings between γ_1 and $\tilde{\gamma}_1$, γ_2 , γ_3 open an energy gap Δ between the high- and low-

energy sectors. This gap will be kept finite throughout the braiding protocol, which is guaranteed by enforcing the condition $\rho_1^2 + \rho_2^2 + \rho_3^2 = 1$. We thus obtain a well-defined ground state sector with two degenerate states in each total parity sector.

We next discuss how to obtain the second line of Eq. (1) by allowing the outer MBSs $\tilde{\gamma}_k$ to partially overlap with the respective inner MBSs γ_k . For $\eta = 0$, γ_k and $\tilde{\gamma}_k$ describe MBSs that are fully localized on separate QDs—Fig. 1(c). In this case, tunnel couplings between the inner QDs of subsystems k and k' will couple only γ_k and $\gamma_{k'}$, as described by the first line of Eq. (1). In contrast, for $0 < \eta \leq 1$, γ_k and $\tilde{\gamma}_k$ describe MBSs that have partial weights on both QDs Fig. 1(d). Therefore, the tunnel coupling will instead couple the operators $\frac{1}{\sqrt{1+\eta}}(\gamma_{k/k'} + i\sqrt{\eta} \tilde{\gamma}_{k/k'})$. This creates additional couplings between $\tilde{\gamma}_k$ and $\tilde{\gamma}_{k'}$ as represented by the second line of Eq. (1), see [46] for a formal derivation. Note that we focus on overlaps which do not introduce a finite energy associated with the parity $i\gamma_k \tilde{\gamma}_k$ [instead described by the third line of Eq. (1), see below]. Furthermore, for simplicity of the analytical solution, we assume that the coefficients of the second and third lines of Eq. (1) (η and λ) can be tuned independently. For a QD-based minimal Kitaev chain, this is supported by findings of Ref. [35], see [46] for details. To facilitate an analytical solution below, we furthermore choose the same MBS overlap η for all MBS pairs. The final result qualitatively agrees with a numerical solution for the general case with an effective η [46]. A commonly used quality measure for MBSs is the Majorana polarization M [24,47–50]. An increase in η directly leads to a decrease of the Majorana polarization on the inner edges, $M = \frac{1-\eta}{1+\eta}$ [46].

In Eq. (1), there are no couplings between next-nearest neighbors ($\propto i\gamma_i \tilde{\gamma}_j$). We assume that these couplings have been turned off by an appropriate tuning of the phases of the couplings, which is always possible [35] (note that phase control has been experimentally implemented in a three-site Kitaev chain [34]). In the setup in Fig. 1(a), this is accomplished by the fluxes Φ_{12} and Φ_{13} .

There is, however, no general way to remove the remaining coupling terms in the second line of Eq. (1), which will break the ground state degeneracy and prevent robust braiding operations [35,37]. To compensate for this degeneracy breaking, we introduce couplings of the outer MBSs $\tilde{\gamma}_{2/3}$ to their inner neighbors $\gamma_{2/3}$, described by the third line of Eq. (1). We choose these to be equal to simplify the analytical solution.

Four signs matter for the protocol performance: the total parity σ , the sign of ρ_1 , and there could, in principle, be a “−” (rather than a “+”) between the two terms in the second and third lines of Eq. (1). A mismatch between the relative signs of lines two and three prevents establishing degeneracy via the correction. In the SM [46], we show that the remaining eight sign combinations reduce to two types of protocols which differ from each other in the size of the gap and intermediate results during the protocol, but give the same final result for the double braid in the adiabatic limit—Eq. (15). For the remainder of the paper, we take $\sigma = -1$ and all other signs positive.

Braiding protocol. We start by considering perfectly isolated MBSs ($\eta = 0$) without corrections ($\lambda = 0$) and show that the coupling-based braiding protocol realizes an exchange of

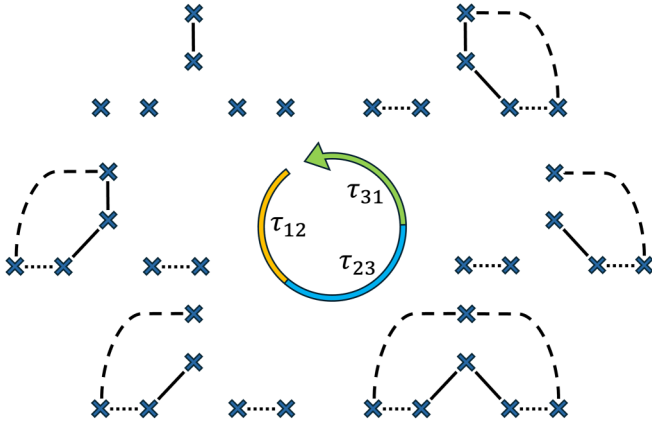


FIG. 2. One cycle of the time-dependent tuning of the couplings in the braiding protocol. The protocol is split into three time intervals τ_{12} (orange), τ_{23} (blue), τ_{31} (green), connecting three coupling configurations as defined in Eqs. (2).

γ_2 and γ_3 by cyclically varying the different MBS couplings [14,15,17,19,35,36,45]. This is done via a closed loop in the three-dimensional parameter space ρ_1, ρ_2, ρ_3 , and we consider the limit where all parameters are changed adiabatically slow compared to the gap Δ , ensuring that the system stays in the ground state sector. The loop in parameter space is split into three separate time intervals $\tau_{12}, \tau_{23}, \tau_{31}$:

$$\begin{aligned} \tau_{12} : \quad & \rho_1 = 1, \rho_2 = 0 \quad \rightarrow \quad \rho_1 = 0, \rho_2 = 1, \\ \tau_{23} : \quad & \rho_2 = 1, \rho_3 = 0 \quad \rightarrow \quad \rho_2 = 0, \rho_3 = 1, \\ \tau_{31} : \quad & \rho_3 = 1, \rho_1 = 0 \quad \rightarrow \quad \rho_3 = 0, \rho_1 = 1. \end{aligned} \quad (2)$$

See the solid lines in Fig. 2. Next will follow a detailed discussion of the first time interval, τ_{12} , including a solution of the implemented rotation for a system hosting isolated MBSs. The other two protocol intervals τ_{23}, τ_{31} work analogously.

We start by coupling MBSs $\gamma_1, \tilde{\gamma}_1$ ($\rho_1 = 1$); see the upper left panel of Fig. 2. The ground state then has $i\gamma_1\tilde{\gamma}_1 = -1$ and, because of the four uncoupled MBSs, is twofold degenerate within each total parity sector. In an experiment, the system should be initialized in one of these ground states (e.g., as described in Ref. [35]).

During the first time interval τ_{12} , the coupling of γ_1 is transferred from $\tilde{\gamma}_1$ to γ_2 by first tuning $\rho_2 > 0$ and then $\rho_1 = 0$; see the left column of Fig. 2. Tuning the couplings in this way guarantees a gap to excited states throughout the whole step. The exact parametrization of that exchange is unimportant as long as the changes are adiabatic.

In general, the result of the adiabatic braiding protocol can be found from the non-Abelian Berry phase (see below). However, for $\eta = 0$ a simple intuitive argument suffices. During the time interval τ_{12} , introducing the coupling to the second MBS, γ_2 , effectively couples γ_1 to a superposition of $\tilde{\gamma}_1$ and γ_2 described by γ_Δ :

$$H(\tau \in \tau_{12})/\Delta = i\gamma_1\gamma_\Delta, \quad \gamma_\Delta = \rho_1\tilde{\gamma}_1 + \rho_2\gamma_2. \quad (3)$$

The definition of γ_Δ required by the diagonalization directly yields an uncoupled MBS

$$\gamma'_\Delta = \rho_1\gamma_2 - \rho_2\tilde{\gamma}_1, \quad (4)$$

which is not part of the diagonalized Hamiltonian (Eq. (3)). The above parametrization of γ'_Δ is fully determined by demanding consistency with the initial conditions and anti-commutation relations

$$\gamma'_\Delta \stackrel{\rho_2=0}{=} \gamma_2, \quad \{\gamma_\Delta, \gamma'_\Delta\} = 0. \quad (5)$$

Accordingly, the first step of the protocol described by Eqs. (3) and (4) implements the transformation $\tilde{\gamma}_1 \rightarrow \gamma_2$, $\gamma_2 \rightarrow -\tilde{\gamma}_1$ while leaving all other operators unchanged. This corresponds to the unitary

$$U_{12} = \exp\left(\frac{\pi}{4} \gamma_2 \tilde{\gamma}_1\right). \quad (6)$$

The same kind of analysis also yields the results of the time intervals τ_{23} and τ_{31} , where the coupling is transferred to γ_3 and finally back to $\tilde{\gamma}_1$. These steps are described by the unitaries

$$U_{23} = \exp\left(\frac{\pi}{4} \gamma_3 \gamma_2\right), \quad U_{31} = \exp\left(\frac{\pi}{4} \tilde{\gamma}_1 \gamma_3\right). \quad (7)$$

Thus, the non-Abelian Berry phase of the complete braiding protocol is

$$U(\eta = 0) = U_{31} \times U_{23} \times U_{12} = \exp\left(\frac{\pi}{4} \gamma_3 \gamma_2\right), \quad (8)$$

implementing an exchange of γ_2 and γ_3 .

Now we consider the case of imperfect MBSs, $\eta > 0$. This does not affect the initialization in the ground state of $i\gamma_1\tilde{\gamma}_1$. However, turning on the coupling $\rho_2 i\gamma_1\gamma_2$ now introduces also couplings between $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$, $\rho_2 \eta i\tilde{\gamma}_1\tilde{\gamma}_2$ [dashed lines in Figs. 1(b) and 2], lifting the ground state degeneracy. This degeneracy is restored via our correction term $\lambda(i\gamma_2\tilde{\gamma}_2 + i\gamma_3\tilde{\gamma}_3)$ [dotted lines in Figs. 1(b) and 2], which has to be tuned at the same time as ρ_1, ρ_2 . The exact relation between λ and ρ_1, ρ_2 needed to maintain ground-state degeneracy will be established below.

We can use the intuition from the $\eta = 0$ result to guess the possible braiding results for the general case. The protocol exchanges MBSs with each other when they are combined into a superposition of the ground-state basis during a protocol interval, like $\gamma_\Delta, \gamma'_\Delta$. This diagonalization always combines pairs of next-nearest neighbors that couple to a shared MBS. Therefore, we can divide all available MBSs into two groups: $\gamma_2, \gamma_3, \tilde{\gamma}_1$ and $\tilde{\gamma}_2, \tilde{\gamma}_3, \gamma_1$. The couplings between the groups can be arbitrary, but there are no direct couplings between two MBSs of the same group. We find that rotations can only consist of exchanges within those groups [46]. Combinations including $\gamma_1, \tilde{\gamma}_1$ can be excluded because they correspond to rotations to excited states of the initial and final configuration ($\rho_1 = 1$). Therefore, the most general operator the protocol can implement must be of the form

$$U(\eta) = \exp(\phi \gamma_3 \gamma_2) \exp(\tilde{\phi} \tilde{\gamma}_3 \tilde{\gamma}_2). \quad (9)$$

Next, we present an analytical solution for $\phi, \tilde{\phi}$, valid for arbitrary η . This solution is split into two steps. First, we diagonalize the Hamiltonian. This enables us to establish degeneracy by tuning λ and to describe the ground-state sector via a projector. This projector is then used to find the time

evolution in the adiabatic limit, which is described by the non-Abelian Berry phase.

Identifying the ground-state MBSs. We start by rewriting the diagonalization of H for isolated MBSs introduced in Eqs. (3) which was valid for the protocol interval τ_{12} . We extend this over the whole protocol time by redefining $\gamma_\Delta = \rho_1 \tilde{\gamma}_1 + \rho_2 \tilde{\gamma}_2 + \rho_3 \tilde{\gamma}_3$, similar to [45]. γ_Δ constitutes the MBS that establishes the gap Δ with γ_1 throughout the protocol, thereby defining the ground-state sector.

Due to the MBS overlap η and correction terms λ , we need to take additional steps to diagonalize the full system of Eq. (1). The full diagonalization describes the system as $H/\Delta^D = i\gamma_1^D \gamma_\Delta^D$, equivalent to Eqs. (3), by rotating $\gamma_1, \tilde{\gamma}_2, \tilde{\gamma}_3$ into γ_1^D and $\tilde{\gamma}_1, \gamma_2, \gamma_3$ into γ_Δ^D . This rotation absorbs the terms of the errors and corrections and expresses their effect on the ground-state sector defined by $i\gamma_1^D \gamma_\Delta^D = -1$ separated from the higher-energy states by $\Delta^D > 0$, see SM [46]. From this diagonalization, we obtain two pieces of information that are necessary to implement the correction and to describe the effective ground-state sector. First, it introduces the ground state projection operator

$$P = \frac{1}{2}(1 - i\gamma_1^D \gamma_\Delta^D). \quad (10)$$

Furthermore, η introduces a finite splitting ε of the ground state sector. We compensate for this splitting by tuning λ . From the diagonalized Hamiltonian, we find the degeneracy condition

$$\varepsilon(\lambda, \eta) - \lambda = 0, \quad (11)$$

which defines the nonlinear dependence of the correction λ on the MBS overlap η , which is furthermore dependent on the protocol parameters ρ_1, ρ_2, ρ_3 and therefore time dependent throughout the protocol. The required correction depends on σ [46], which prevents using this protocol for actual topological quantum computing, where we need to perform gates on the qubits without knowing their state. There is no obvious, general analytical solution to Eq. (11), but we find in the calculation of the Berry phase that it only requires the solution of this expression in the second protocol interval τ_{23} , defined by $\rho_1 = 0$, where it simplifies crucially—see [46]. In an experiment, the degeneracy condition in Eq. (11) needs to be determined via, for example, tunneling spectroscopy at each protocol step. A quasistatic deviation $\delta\lambda$ would lead to a residual energy splitting $\propto \delta\lambda \Delta$. The protocol duration then has to be short compared to the inverse of this splitting to avoid a dynamical phase.

Calculating the Berry phase. As the next step, we use Eq. (10) to obtain the braiding operator in the ground-state sector from the non-Abelian Berry phase [51,52]

$$U(\eta) = \mathcal{T} \exp \left(- \oint [P, \partial_\omega P] d\omega \right), \quad (12)$$

where $\mathcal{T}$ is the time-ordered path integral. The Berry phase describes the induced ground-state rotation by integrating the effect of the ground-state projector evolution over the whole protocol loop. The integrand is, in general, noncommuting with itself at different protocol times, making the time-ordered integration nontrivial. We simplify the calculation by first splitting it into the three protocol intervals defined in Eqs. (2) $U(\eta) = U_{31} \cdot U_{23} \cdot U_{12}$. Afterward, we solve the time-ordered

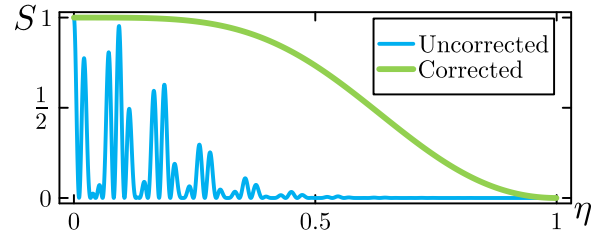


FIG. 3. MBS similarity S for a double braid as a function of the MBS overlap η . Green line: analytical result given by Eq. (15) for a perfectly corrected protocol. Blue oscillating line: Numerical result for the uncorrected protocol (details in Sec. VII. in the SM [46]).

integration of each interval separately and multiply the operator exponentials for all time-steps—see [46]. The solution yields the analytical expressions for the angles $\phi(\eta), \tilde{\phi}(\eta)$ in Eq. (9):

$$\begin{aligned} \phi &= \frac{\pi}{4} \left(1 - \frac{\eta^2}{\sqrt{1 + \eta^2 + \eta^4}} \right), \\ \tilde{\phi} &= \frac{\pi}{4} \left(1 - \frac{1}{\sqrt{1 + \eta^2 + \eta^4}} \right). \end{aligned} \quad (13)$$

Note that these angles are only correct when acting with $U(\eta)$ on the ground-state sector.

Discussion. The result of the braiding protocol in the fully adiabatic limit for arbitrary MBS overlaps η and perfect corrections λ is given by Eq. (9) together with Eq. (13). We can further simplify the solution using conservation of total parity, $\tilde{\gamma}_3 \tilde{\gamma}_2 = -\gamma_3 \gamma_2$, yielding

$$U(\eta) = \exp((\phi - \tilde{\phi}) \gamma_3 \gamma_2). \quad (14)$$

This representation highlights that the braiding protocol implements a partial exchange of γ_2, γ_3 given by the angle $\phi_{\text{eff}}(\eta) = \phi - \tilde{\phi}$, interpolating between a full exchange for isolated MBSs $\phi_{\text{eff}}(\eta = 0) = \frac{\pi}{4}$ and a trivial exchange for a fermionic system $\phi_{\text{eff}}(\eta = 1) = 0$.

Next, we introduce the MBS similarity $S(U)$ that quantifies how much the braiding result resembles that of fully isolated MBSs. We base $S(U)$ on a double braid, i.e., on U^2 , which provides a clearer and more sensitive experimental signature than a single braid [19,35]. From Eq. (14), we find the simple result

$$S(U) = \text{Tr}\{U^2(\eta)^\dagger U^2(0)\}^2/4 = \sin^2 \left(\frac{\pi}{2} \sqrt{\frac{(1 - \eta^2)^3}{(1 - \eta^6)}} \right), \quad (15)$$

which quantifies the coupling between the unitary transformation $U^2(\eta)$ implemented by the double braid protocol and the expected result for an isolated MBS system, $U^2(\eta = 0) = \exp(\frac{\pi}{2} \gamma_3 \gamma_2)$ —see green line in Fig. 3. For $\eta \lesssim 0.25$, the result very closely resembles the braiding of isolated MBSs. For larger values, the fermionic character becomes increasingly evident and the result approaches the trivial fermion result $U^2(\eta = 1) = 1$. However, the result remains non-Abelian for all $\eta < 1$.

We compare this to the result of the braiding protocol without corrections ($\lambda = 0$), which leads to the transformation U_{uncorr} , suffering from breaking of the ground-state degeneracy. The result is shown in Fig. 3 (blue line), which is based on numerically solving the time-dependent Schrödinger equation, see Sec. VII. in the SM [46] for a description of the method as well as the parameters used and Ref. [53] for the code. The uncorrected curve shows MBS behavior in a very small region around $\eta \approx 0$. For larger MBS overlaps, the dynamical phase leads to an additional rotation within the almost degenerate ground state sector [35,37]. For even larger MBS overlaps, $\eta \gtrsim 0.25$, the energy splitting within the ground state sector becomes so large that the protocol acts adiabatically concerning the splitting and the system remains in the initial state. While these three regimes are universal, the position of boundaries between them along η depends on the detailed protocol parametrization and relation between the induced energy splitting and the speed of parameter changes.

$S(U)$ is not easily accessible in experiments. However, it is closely related to the result of an experiment where (i) the MBS pairs $\gamma_2, \tilde{\gamma}_2$ and $\gamma_3, \tilde{\gamma}_3$ are initialized in a given parity state, (ii) the double braid is performed, and (iii) the parities of the same pairs are measured again. The result for fully isolated MBS is a parity flip, while the trivial fermion result is no change compared to the initialized parities.

Conclusions. In this Letter, we have presented a generalized coupling-based braiding protocol applied to a model that

interpolates between isolated MBSs and conventional zero-energy fermions, the latter being equivalent to spatially fully overlapping MBSs. The MBS overlaps introduce a ground state splitting during the protocol, which we showed can be corrected for by additional MBS couplings. Except in the perfect fermion limit, the braiding result is non-Abelian and robust to changes in protocol details such as the strength and duration of MBS couplings. Our work expands the notion of what constitutes a non-Abelian state, with possible impacts ranging from topological quantum computing protocols to classification and experimental detection of topological phases. We believe that quantum-dot-based minimal Kitaev chains are an ideal platform to realize our protocol, as they allow control of all required parameters.

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Data availability. The data that support the findings of this article are openly available [53].

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- [1] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, Non-Abelian anyons and topological quantum computation, *Rev. Mod. Phys.* **80**, 1083 (2008).
 - [2] C. W. J. Beenakker, Search for non-Abelian Majorana braiding statistics in superconductors, *SciPost Phys. Lect. Notes* **15** (2020).
 - [3] A. Yazdani, F. von Oppen, B. I. Halperin, and A. Yacoby, Hunting for Majoranas, *Science* **380**, eade0850 (2023).
 - [4] S. Das Sarma, M. Freedman, and C. Nayak, Majorana zero modes and topological quantum computation, *npj Quantum Inf.* **1**, 15001 (2015).
 - [5] T. Karzig, *et al.*, Scalable designs for quasiparticle-poisoning-protected topological quantum computation with Majorana zero modes, *Phys. Rev. B* **95**, 235305 (2017).
 - [6] P. Marra, Majorana nanowires for topological quantum computation, *J. Appl. Phys.* **132**, 231101 (2022).
 - [7] F. Wilczek, Majorana returns, *Nat. Phys.* **5**, 614 (2009).
 - [8] J. Alicea, New directions in the pursuit of Majorana fermions in solid state systems, *Rep. Prog. Phys.* **75**, 076501 (2012).
 - [9] M. Leijnse and K. Flensberg, Introduction to topological superconductivity and Majorana fermions, *Semicond. Sci. Technol.* **27**, 124003 (2012).
 - [10] C. Beenakker, Search for Majorana fermions in superconductors, *Annu. Rev. Condens. Matter Phys.* **4**, 113 (2013).
 - [11] D. A. Ivanov, Non-Abelian statistics of half-quantum vortices in p -wave superconductors, *Phys. Rev. Lett.* **86**, 268 (2001).
 - [12] J. Alicea, Y. Oreg, G. Refael, F. von Oppen, and M. P. A. Fisher, Non-Abelian statistics and topological quantum information processing in 1D wire networks, *Nat. Phys.* **7**, 412 (2011).
 - [13] K. Flensberg, Non-Abelian operations on Majorana fermions via single-charge control, *Phys. Rev. Lett.* **106**, 090503 (2011).
 - [14] J. D. Sau, D. J. Clarke, and S. Tewari, Controlling non-Abelian statistics of Majorana fermions in semiconductor nanowires, *Phys. Rev. B* **84**, 094505 (2011).
 - [15] B. van Heck, A. R. Akhmerov, F. Hassler, M. Burrello, and C. W. J. Beenakker, Coulomb-assisted braiding of Majorana fermions in a Josephson junction array, *New J. Phys.* **14**, 035019 (2012).
 - [16] D. Aasen *et al.*, Milestones toward Majorana-based quantum computing, *Phys. Rev. X* **6**, 031016 (2016).
 - [17] M. Hell, M. Leijnse, and K. Flensberg, Two-dimensional platform for networks of Majorana bound states, *Phys. Rev. Lett.* **118**, 107701 (2017).
 - [18] S. Plugge, A. Rasmussen, R. Egger, and K. Flensberg, Majorana box qubits, *New J. Phys.* **19**, 012001 (2017).
 - [19] D. J. Clarke, J. D. Sau, and S. Das Sarma, Probability and braiding statistics in Majorana nanowires, *Phys. Rev. B* **95**, 155451 (2017).
 - [20] A. Y. Kitaev, Unpaired Majorana fermions in quantum wires, *Phys. Usp.* **44**, 131 (2001).
 - [21] J. D. Sau and S. Das Sarma, Realizing a robust practical Majorana chain in a quantum-dot-superconductor linear array, *Nat. Commun.* **3**, 964 (2012).
 - [22] M. Leijnse and K. Flensberg, Parity qubits and poor man's Majorana bound states in double quantum dots, *Phys. Rev. B* **86**, 134528 (2012).

- [23] I. C. Fulga, A. Haim, A. R. Akhmerov, and Y. Oreg, Adaptive tuning of Majorana fermions in a quantum dot chain, *New J. Phys.* **15**, 045020 (2013).
- [24] A. Tsintzis, R. S. Souto, and M. Leijnse, Creating and detecting poor man's Majorana bound states in interacting quantum dots, *Phys. Rev. B* **106**, L201404 (2022).
- [25] C.-X. Liu, G. Wang, T. Dvir, and M. Wimmer, Tunable superconducting coupling of quantum dots via Andreev bound states in semiconductor-superconductor nanowires, *Phys. Rev. Lett.* **129**, 267701 (2022).
- [26] R. Seoane Souto and R. Aguado, Subgap states in semiconductor-superconductor devices for quantum technologies: Andreev qubits and minimal Majorana chains, in *New Trends and Platforms for Quantum Technologies*, edited by R. Aguado, R. Citro, M. Lewenstein, and M. Stern (Springer Nature Switzerland, Cham, 2024), p. 133.
- [27] G. Wang, T. Dvir, G. P. Mazur, C.-X. Liu, N. van Loo, S. L. D. ten Haaf, A. Bordin, S. Gazibegovic, G. Badawy, E. P. A. M. Bakkers, M. Wimmer, and L. P. Kouwenhoven, Singlet and triplet Cooper pair splitting in hybrid superconducting nanowires, *Nature (London)* **612**, 448 (2022).
- [28] T. Dvir, G. Wang, N. van Loo, C.-X. Liu, G. P. Mazur, A. Bordin, S. L. D. ten Haaf, J.-Y. Wang, D. van Driel, F. Zlatelli, X. Li, F. K. Malinowski, S. Gazibegovic, G. Badawy, E. P. A. M. Bakkers, M. Wimmer, and L. P. Kouwenhoven, Realization of a minimal Kitaev chain in coupled quantum dots, *Nature (London)* **614**, 445 (2023).
- [29] Q. Wang, S. L. D. ten Haaf, I. Kulesh, D. Xiao, C. Thomas, M. J. Manfra, and S. Goswami, Triplet correlations in Cooper pair splitters realized in a two-dimensional electron gas, *Nat. Commun.* **14**, 4876 (2023).
- [30] A. Bordin, G. Wang, C.-X. Liu, S. L. D. ten Haaf, N. van Loo, G. P. Mazur, D. Xu, D. van Driel, F. Zlatelli, S. Gazibegovic, G. Badawy, E. P. A. M. Bakkers, M. Wimmer, L. P. Kouwenhoven, and T. Dvir, Tunable crossed Andreev reflection and elastic cotunneling in hybrid nanowires, *Phys. Rev. X* **13**, 031031 (2023).
- [31] D. van Driel, R. Koch, V. P. M. Sietses, S. L. D. ten Haaf, C.-X. Liu, F. Zlatelli, B. Roovers, A. Bordin, N. van Loo, G. Wang, J. C. Wolff, G. P. Mazur, T. Dvir, I. Kulesh, Q. Wang, A. M. Bozkurt, S. Gazibegovic, G. Badawy, E. P. A. M. Bakkers, M. Wimmer, *et al.*, Cross-platform autonomous control of minimal Kitaev chains, [arXiv:2405.04596](https://arxiv.org/abs/2405.04596).
- [32] A. Bordin, X. Li, D. van Driel, J. C. Wolff, Q. Wang, S. L. D. ten Haaf, G. Wang, N. van Loo, L. P. Kouwenhoven, and T. Dvir, Crossed Andreev reflection and elastic cotunneling in three quantum dots coupled by superconductors, *Phys. Rev. Lett.* **132**, 056602 (2024).
- [33] S. L. D. ten Haaf, Q. Wang, A. M. Bozkurt, C.-X. Liu, I. Kulesh, P. Kim, D. Xiao, C. Thomas, M. J. Manfra, T. Dvir, M. Wimmer, and S. Goswami, A two-site Kitaev chain in a two-dimensional electron gas, *Nature (London)* **630**, 329 (2024).
- [34] A. Bordin, F. J. B. Everts, B. Roovers, J. D. T. Luna, W. D. Huisman, F. Zlatelli, G. P. Mazur, S. L. D. ten Haaf, G. Badawy, E. P. A. M. Bakkers, C.-X. Liu, R. S. Souto, N. van Loo, and L. P. Kouwenhoven, Probing Majorana localization of a phase-controlled three-site Kitaev chain with an additional quantum dot, [arXiv:2504.13702](https://arxiv.org/abs/2504.13702).
- [35] A. Tsintzis, R. S. Souto, K. Flensberg, J. Danon, and M. Leijnse, Majorana qubits and non-Abelian physics in quantum dot-based minimal Kitaev chains, *PRX Quantum* **5**, 010323 (2024).
- [36] S. Miles, F. Zlatelli, A. M. Bozkurt, M. Wimmer, and C.-X. Liu, Braiding Majoranas in a linear quantum dot-superconductor array: Mitigating the errors from Coulomb repulsion and residual tunneling, *Phys. Rev. B* **113**, 085302 (2026).
- [37] T. Huang, R. Zhang, X. Li, X.-J. Liu, X. C. Xie, and Y. Wu, Unified model for non-Abelian braiding of Majorana and Dirac fermion zero modes, *Sci. China: Phys., Mech. Astron.* **69**, 227211 (2026).
- [38] Y. Zhang, Y. Wu, J. Liu, and X. C. Xie, Quantifying the non-Abelian property of Andreev bound states in inhomogeneous Majorana nanowires, *Phys. Rev. B* **111**, 205411 (2025).
- [39] P. Boross and A. Pályi, Braiding-based quantum control of a Majorana qubit built from quantum dots, *Phys. Rev. B* **109**, 125410 (2024).
- [40] A. Vuik, B. Nijholt, A. R. Akhmerov, and M. Wimmer, Reproducing topological properties with quasi-Majorana states, *SciPost Phys.* **7**, 061 (2019).
- [41] D. J. Clarke, Experimentally accessible topological quality factor for wires with zero energy modes, *Phys. Rev. B* **96**, 201109(R) (2017).
- [42] E. Prada, R. Aguado, and P. San-Jose, Measuring Majorana nonlocality and spin structure with a quantum dot, *Phys. Rev. B* **96**, 085418 (2017).
- [43] M. T. Deng, S. Vaitiekėnas, E. B. Hansen, J. Danon, M. Leijnse, K. Flensberg, J. Nygård, P. Krogstrup, and C. M. Marcus, Majorana bound state in a coupled quantum-dot hybrid-nanowire system, *Science* **354**, 1557 (2016).
- [44] M.-T. Deng, S. Vaitiekėnas, E. Prada, P. San-Jose, J. Nygård, P. Krogstrup, R. Aguado, and C. M. Marcus, Nonlocality of Majorana modes in hybrid nanowires, *Phys. Rev. B* **98**, 085125 (2018).
- [45] T. Karzig, F. Pientka, G. Refael, and F. von Oppen, Shortcuts to non-Abelian braiding, *Phys. Rev. B* **91**, 201102(R) (2015).
- [46] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/3vcb-pxp6> for the full analytical derivation of all results and additional numerical investigations.
- [47] N. Sedlmayr and C. Bena, Visualizing Majorana bound states in one and two dimensions using the generalized Majorana polarization, *Phys. Rev. B* **92**, 115115 (2015).
- [48] N. Sedlmayr, J. M. Aguiar-Hualde, and C. Bena, Majorana bound states in open quasi-one-dimensional and two-dimensional systems with transverse Rashba coupling, *Phys. Rev. B* **93**, 155425 (2016).
- [49] S. V. Aksenov, A. O. Zlotnikov, and M. S. Shustin, Strong Coulomb interactions in the problem of Majorana modes in a wire of the nontrivial topological class BDI, *Phys. Rev. B* **101**, 125431 (2020).
- [50] W. Samuelson, V. Svensson, and M. Leijnse, Minimal quantum dot based Kitaev chain with only local superconducting proximity effect, *Phys. Rev. B* **109**, 035415 (2024).
- [51] J. E. Avron, L. Sadun, J. Segert, and B. Simon, Chern numbers, quaternions, and Berry's phases in Fermi systems, *Commun. Math. Phys.* **124**, 595 (1989).
- [52] F. Hassler, Topological quantum computing, [arXiv:2410.13547](https://arxiv.org/abs/2410.13547).
- [53] M. Nitsch, V. Svensson, W. Samuelson, K. Nestmann, J. Danon, K. Flensberg, R. Seoane Souto, and M. Leijnse, Code for the article "Adiabatic nonabelian braiding of imperfect Majoranas" (v1-arXiv), Zenodo (2025), <https://doi.org/10.5281/zenodo.15879397>.